

TOLUPE MATTHEW BUSOYE

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Research Interests: TinyML & Edge AI • Embedded Systems • Cyber-Physical Systems • Intelligent Energy Systems • Renewable Energy • Internet of Things • Real-Time Systems

Education

Master of Science (M.Sc.) in Mathematical Sciences (AI for Science Stream) 2026 – 2027
African Institute for Mathematical Sciences (AIMS) South Africa / Google DeepMind

- Theoretical knowledge and practical skills in mathematics, machine learning, and AI for scientific applications.

B.Sc. Electrical and Electronic Engineering 2019 – 2025
University of Ibadan, Nigeria CGPA: 3.94/4.00 — First Class

- Best Graduating Student of the Department and Faculty of Technology
- Studied electronic circuits, microprocessors and embedded systems, electric machines, programming, and cross-disciplinary engineering courses
- **Thesis:** Development of a Commercialised Solar-Based Mobile Energy Hub

Senior Secondary Certificate Examination 2015 – 2017
Pacesetters Comprehensive College

- Ranked among the top 5%; received the best student award in Physics, Mathematics, and Economics

Research Experience

TinyML & Edge AI Lead, ML Collective Apr 2026 – Present
Mentored by Steven Kolawole and independent researchers within the ML Collective network. Model optimisation and deployment on microcontrollers for real-world applications; focus on energy-efficient embedded inference.

Hardware & TinyML Researcher, Powerstart Integrated Technologies Limited (PowerLabs) 2025 – 2026
Supervised by Dr. Eigbe Egbonbon-aye. Optimised TinyML models for deployment on microcontrollers and embedded systems with as little as 2kB of available memory. Developed deep microcompression pipeline combining structured pruning and bit-packed quantisation; work accepted to ICML 2026 Global South in AI Workshop.

Lead Student Investigator, University of Ibadan 2024 – 2025
Supervised by Prof. O. O. Olakanmi. Developed a commercialised solar-based mobile energy hub for off-grid communities. Designed a full-stack hardware system: from circuit schematics and veroboard prototyping to embedded C/C++ firmware and a companion Flutter/Dart mobile app.

Publications and Articles

Conference Papers

- T. M. Busoye, Q. A. Jokomba, O. M. Busoye, A. O. Joledo. **Edge-Deployed Machine Learning Sensor (MLSensor) for Real-Time Anomaly Detection in Electrical Submersible Systems.** *SPE Nigeria Annual International Conference and Exhibition, 2026.* DOI: [10.2118/235027-MS](https://doi.org/10.2118/235027-MS)
- O. M. Busoye, T. M. Busoye, E. Eigbe. **Deep Microcompression: Structured Pruning and Bit-packed Quantization for Microcontrollers.** *PMLR Vol. 306 — ICML 2026 Global South in AI Workshop.* DOI: [10.48550/arXiv.2609.05081](https://doi.org/10.48550/arXiv.2609.05081)

Manuscripts Under Review / In Preparation

- O. M. Busoye, T. M. Busoye, E. Eigbe. (in preparation). **Right In-Place Convolution: A Simple, General, and Near-Optimal Strategy for Memory-Efficient CNN Inference.**
- T. M. Busoye. (in preparation). **A Compile-Time and Memory-Efficient Approach to SunSpec Models with FlatBuffers and Direct Modbus Register Mapping.**

Thesis & Articles

- T. M. Busoye. *Development of a Commercialised Solar-Based Mobile Energy Hub.* Undergraduate thesis, University of Ibadan, 2025. DOI: [10.13140/RG.2.2.10441.86886](https://doi.org/10.13140/RG.2.2.10441.86886)
- T. M. Busoye. Inside TensorFlow Lite Micro on RP2040/RP2350: Setup and Project Structure Explained. *Medium*, Sep 2025. [Link](#)

Industrial Experience

Embedded Systems Engineer Jun 2025 – Present
Powerstart Integrated Technologies Limited (PowerLabs), Lagos

- Led the development of the PAI mobility battery charging station for energy trading.
- Implemented SunSpec standard into PAI V2, enabling seamless integration with over 10 different DER devices.
- Ported existing firmware to Zephyr RTOS and developed a custom Zephyr board support package (BSP) for PowerLabs hardware.
- Developed a hardware testing framework and integration-testing workflow supporting test-driven development
- Create and maintain a coding style guide to standardise hardware-related software development

Embedded Systems Intern

Jan 2024 – Jun 2024

Powerstart Integrated Technologies Limited (PowerLabs), Lagos

- Designed and developed a PAI V1 device for power event capture based on Raspberry Pi Pico MCU
- Collaborated on software components for transmitting power events to Amazon AWS cloud endpoints
- Designed schematics and PCBs for Power Stool, a Bluetooth-controllable desk

Hardware & Electronics Intern

Jun 2023 – Sep 2023

Reeddi Technologies, Lagos

- Repaired 50 Reeddi Capsules, Energy Boxes, and Big Energy devices
- Designed a charge station for 100 Reeddi Capsules and set up Big Energy installations
- Developed Reeddi Capsule tracking systems

STEM Tutor & Engineer

2021 – 2024

Switches Technology, University of Ibadan

- Tutored and trained over 10 students in Arduino, Raspberry Pi, and embedded systems
- Designed innovative projects in embedded systems, IoT, and AI—including a Smart Distribution Board, Smart Bin, and Responsive Prosthetic Leg
- Oversaw 3D printing projects including a Head Mount Mouse and a 6-DOF Robotic Arm

Teaching Experience

Academic Tutor

2021 – 2025

Faculty of Technology / Toppers Tutorial / MFMCF Fellowship, University of Ibadan

- Tutored over 270 students across Elementary Computer Programming (FORTRAN), PHY103, and MAT111/MAT121
- Prepared academic materials and organised tutorials for colleagues and junior students

Leadership and Volunteering

Lead, ML Systems Study group

2026

ML Collective

- Managed over 20 students to understand the foundational principles of machine learning systems using the cs249r_book by Harvard University.
- Led weekly study sessions where we discussed each week's chapter and programming module implementation.

President (2023/24) & PRO (2021–2023), Society of Electrical & Electronic Engineering Students (SEEES)

2021 – 2024

University of Ibadan

- Responsible for over 270 electrical and electronic engineering students as President
- Organised an industrial visit to Egbin Power Plc, Nigeria's largest power-generating station
- Coordinated workshops bridging theoretical knowledge and industry-required practical skills
- Increased financial resources for the next administration by over 2,000%
- Led publicity and communication for all society activities as PRO; served as official spokesperson

Lead & Co-Lead, Hardware & IoT, Google Developer Student Club (GDSC)

2021/22 – 2022/23

University of Ibadan

- Organised technical sessions and workshops on hardware, IoT, and embedded systems for over 50 students.
- Served as Technical Lead, TechQuest GDSC UI 2023

Student Representative Council Member, 10th Assembly

2022/2023

Student Union, University of Ibadan

- Represented 100 students in constituency; responsible for policy-making
- Member, UISU House Committee on Audit and Electoral Matters

Scholarships and Awards

- **3rd Place—Africa Energy Challenge, SPE Africa Annual Student Congress**
Pan-African student energy-innovation competition hosted by the Society of Petroleum Engineers.

Apr 2025

- **1st Place—SPE Energy Challenge, Student Technical Symposium & Exhibition, Nigeria** Mar 2024
National student energy-engineering competition among Nigerian universities, organised by the Society of Petroleum Engineers, Nigeria.
- **NNPC/TotalEnergies National Merit Scholarship** 2021 – 2024
Competitive national scholarship for top-performing Nigerian STEM undergraduates.
- **Federal Scholarship Board (FSB), Nigeria** 2021 – 2023
Nigerian federal government merit scholarship for outstanding undergraduates.
- **SPE Lagos Section Male Undergraduate Scholarship (Engineering)** 2023
Merit scholarship awarded by the SPE Lagos Section to top engineering undergraduates.
- **Agbami Medical and Engineering Professionals Scholarship Award** 2019 – 2022
Merit-based scholarship for Nigerian engineering and medical students.
- **Most Outstanding in Academics—University of Ibadan Student Union** 2023/24
Top academic distinction among the University's undergraduate student body.
- **5× Dean's Honour List—Faculty of Technology** 2018/19 – 2023/24
Awarded each session to the top-performing students in the Faculty.
- **Most Outstanding Member—Junior Chamber International Nigeria, UI** 2022/24
Recognises leadership and service contribution within the chapter.
- **4× Academic Excellence Awards—MFMCF UI** 2020/21 – 2023/24
Annual academic distinction awarded within the fellowship.

Workshops and Conferences

- Under the Hood with Zephyr: Bringing up the STM32G473 Board, Powerstart Integrated Technologies Sep 2026
- Adapting to Change, Edinburgh Earth Initiative, University of Edinburgh Nov 2024
- Nigeria Annual International Conference and Exhibition, SPE Nigeria Aug 2025
- Student Professional Awareness Conference, IEEE, University of Ibadan Jul 2023
- Exhibitor, Energy Searchlight 2.0, Energy Club, University of Ibadan Feb 2025
- Exhibitor, Ambizero Conference, Energy Club & SPE, University of Ibadan Jun 2024
- Artificial Intelligence and Clean Energy (AI4CE) Workshop, First Technical University 2021

Selected Projects

- **IoT Air Quality Monitoring System** — End-to-end real-time monitoring solution with cloud storage and mobile/desktop dashboard; awarded £1,500 grant by the Edinburgh Earth Initiative. [GitHub](#)
- **Edge-Based Smart Meter** — Offline metering system for local energy data acquisition and TinyML-based analytics on constrained embedded hardware. [GitHub](#)
- **Power Leaf** — Pay-per-use solar energy platform integrating IoT monitoring, prepaid billing, and mobile control for scalable renewable-energy access. [GitHub](#)
- **Automated Solar Power Trading and Distribution Unit** — Smart solar-power distribution prototype with priority-based energy arbitration; 1st place, SPE Energy Challenge 2024
- **Other projects:** Hybrid solar/grid charging station (100 Reeddi Capsules), STM32 PCB design & fabrication, Smart Desk/Power Stool PCB, solar-powered traffic light, smart energy distribution board, Bluetooth programmable display, responsive prosthetic leg
- Over 15 solar and inverter installations across residential and commercial sites

Skills and Languages

- Technical:** Circuit design, PCB design, Solar and inverter installation, Electronic prototyping
- Software:** C/C++, Python, Zephyr, Linux, Buildroot, Pytorch, KiCad, Docker, Git, Flutter/Dart, L^AT_EX, Microsoft Office, Canva
- Strengths:** Fast learning, Communication, Leadership, Time management, Adaptability, Creativity
- Languages:** English (fluent), Yoruba (native)

Professional Affiliations

Graduate Member, Nigerian Society of Engineers (GMNSE) · Student Member, Society of Petroleum Engineers (SPE) · IEEE · SEEES · Energy Club, UI · Google Developer Student Club (GDSC) · Junior Chamber International Nigeria (JCIN)